

SOC Homelab using Wazuh SIEM, Sysmon and Shuffle SOAR

Project Overview

This project demonstrates the design and implementation of a Security Operations Center (SOC) homelab using Wazuh SIEM, Sysmon, Windows, Kali Linux, and Shuffle SOAR. The lab was built in VMware to simulate real-world attack scenarios, monitor security events, detect threats, and automate security workflows.

Objectives

- Deploy and configure Wazuh SIEM/XDR platform
- Monitor Windows and Linux endpoints
- Collect and analyze security logs
- Detect suspicious activities and attacks
- Integrate Sysmon for advanced endpoint visibility
- Build SOAR workflows using Shuffle
- Perform threat detection and incident investigation

Lab Architecture

Kali Linux (Attacker) | v Windows Endpoint + Sysmon | v Wazuh Agent | v Wazuh Manager | v Wazuh Indexer | v Wazuh Dashboard | v Shuffle SOAR

Technologies Used

- Wazuh SIEM/XDR
- Sysmon
- Shuffle SOAR
- Docker
- VMware Workstation
- Kali Linux
- Windows
- Ubuntu Server
- GitHub

Environment Setup

Wazuh Components

- Wazuh Manager
- Wazuh Indexer
- Wazuh Dashboard

Endpoints

- Windows Endpoint with Wazuh Agent
- Sysmon installed on Windows
- Kali Linux for attack simulation

SOAR Platform

- Shuffle installed using Docker on Ubuntu Server

Attack Simulations

Network Reconnaissance

- Nmap TCP Scan
- Nmap Service Detection
- Nmap OS Detection

Authentication Attacks

- Multiple Failed Login Attempts
- Brute Force Simulation

Endpoint Monitoring

- PowerShell Execution
- Command Prompt Activity
- Process Creation Monitoring
- Registry Modification Monitoring

File Integrity Monitoring

- File Creation
- File Modification
- Unauthorized Changes Detection

Wazuh Features Implemented

Log Collection

- Windows Event Logs
- Sysmon Logs
- Security Logs

Threat Detection

- Suspicious Process Detection
- Authentication Monitoring
- File Integrity Monitoring
- MITRE ATT&CK Mapping

Security Monitoring

- Endpoint Monitoring
- Log Analysis
- Alert Correlation
- Threat Hunting

Shuffle SOAR Integration

Workflow

Attack ↓ Wazuh Alert ↓ Shuffle Webhook ↓ Automated Workflow Execution

Purpose

- Receive alerts from Wazuh
- Process security events
- Automate incident response tasks
- Reduce manual analyst workload

Key Skills Demonstrated

- Security Monitoring
- SIEM Operations
- Threat Detection
- Incident Investigation
- Log Analysis
- Windows Security Monitoring
- Linux Security Monitoring
- SOC Operations
- SOAR Automation
- Docker Deployment

Learning Outcomes

- Understood SIEM architecture and log management
- Learned endpoint monitoring using Sysmon
- Performed attack simulations using Kali Linux
- Investigated security alerts in Wazuh
- Built automation workflows using Shuffle
- Improved incident response and threat hunting skills

Future Enhancements

- TheHive Integration
- VirusTotal Integration
- Suricata IDS Integration
- Active Response Automation
- Threat Intelligence Enrichment
- Email and Slack Notifications

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